

LAB SHEET



K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B.TECH CSE (CORE)

Database & Management System Lab (ENCS254)

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LAB SEET – 2

Objective: To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

• DDL Questions (Data Definition Language)

1. Create table CLASS

```
1 • CREATE DATABASE LAB_SHEET_2;
2 • USE LAB_SHEET_2;
3 • CREATE TABLE CLASS (
4     ID INT PRIMARY KEY,
5     Name VARCHAR(20)
6 );
7 • INSERT INTO CLASS VALUES (1,'Bravo');
8 • INSERT INTO CLASS VALUES (2,'Alex');
9 • INSERT INTO CLASS VALUES (4,'Cheng');
10
```

2. Create table EMPLOYEES

```
• CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(100),
    Phone_Number VARCHAR(20),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary INT,
    Commission_Pct INT,
    Manager_Id INT,
    Department_Id INT
);
• INSERT INTO EMPLOYEES VALUES
(101,'Neha','Sharma','neha@gmail.com','9876543210','2023-01-10','HR',5000,NULL,100,60);
• INSERT INTO EMPLOYEES VALUES
(102,'Rohan','AUSTIN','rohan@gmail.com','9876543211','2022-03-15','IT',7000,NULL,100,70);
• INSERT INTO EMPLOYEES VALUES
(103,'Aman','Verma','aman@gmail.com','9876543212','2021-07-20','Sales',4500,NULL,101,80);
• INSERT INTO EMPLOYEES VALUES
(104,'Priya','Kapoor','priya@gmail.com','9876543213','2020-11-25','Finance',4800,NULL,102,60);
• INSERT INTO EMPLOYEES VALUES
(105,'Karan','Mehta','karan@gmail.com','9876543214','2019-05-30','IT',5500,NULL,100,80);
```

3. Create table Client_master

- ```
CREATE TABLE Client_master (
 ClientNO VARCHAR(10) PRIMARY KEY,
 Name VARCHAR(50),
 Address VARCHAR(100),
 City VARCHAR(50),
 Pincode INT,
 State VARCHAR(50),
 Bal_due INT
);
```

4. Rename table Client\_master to Client12

```
RENAME TABLE Client_master TO Client12;
```

5. Display bal\_due column heading as "BALANCE"

```
SELECT Bal_due AS "BALANCE"
FROM Client_master;
```

## • DML Questions (Data Manipulation Language) ○ Part A: Transactions and Savepoints (CLASS Table)

1. COMMIT

- ```
INSERT INTO CLASS VALUES (5, 'Rahul');
```
- ```
COMMIT;
```

2. SAVEPOINT A

```
UPDATE CLASS SET Name = 'Abhijeet' WHERE ID = 5;
SAVEPOINT A;
```

3. SAVEPOINT B

- ```
INSERT INTO CLASS VALUES (6, 'Chris');
```
- ```
SAVEPOINT B;
```

4. SAVEPOINT C

- ```
INSERT INTO CLASS VALUES (7, 'Bravo');
```
- ```
SAVEPOINT C;
```

5. SELECT ALL FROM CLASS

- ```
SELECT * FROM CLASS;
```

Output:

Result Grid		
	ID	Name
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Abhijeet
	6	Chris
	7	Bravo
•	NULL	NULL

6.ROLLBACK TO B AND SELECT ALL FROM CLASS

- `ROLLBACK TO B;`
- `SELECT * FROM CLASS;`

Output:

Result Grid		
	ID	Name
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Abhijeet
	6	Chris
•	NULL	NULL

8.ROLLBACK TO A AND SELECT ALL FROM CLASS

- `ROLLBACK TO A;`
- `SELECT * FROM CLASS;`

Output:

Result Grid		
	ID	Name
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Abhijeet
•	NULL	NULL

○ Part B: Retrieving Records (EMPLOYEES Table)

1. Employee id, names and salaries

- ```
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

OUTPUT:

|   | Employee_Id | First_Name | Last_Name | Salary |
|---|-------------|------------|-----------|--------|
| ▶ | 101         | Neha       | Sharma    | 5000   |
|   | 102         | Rohan      | AUSTIN    | 7000   |
|   | 103         | Aman       | Verma     | 4500   |
|   | 104         | Priya      | Kapoor    | 4800   |
|   | 105         | Karan      | Mehta     | 5500   |
| * | NULL        | NULL       | NULL      | NULL   |

### 2. Employees under manager 100

- ```
SELECT *
FROM EMPLOYEES
WHERE Manager_Id = 100;
```

Output:

	Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
▶	101	Neha	Sharma	neha@gmail.com	9876543210	2023-01-10	HR	5000	NULL	100	60
	102	Rohan	AUSTIN	rohan@gmail.com	9876543211	2022-03-15	IT	7000	NULL	100	70
	105	Karan	Mehta	karan@gmail.com	9876543214	2019-05-30	IT	5500	NULL	100	80
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Salary $\geq$ 4800

- ```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

Output:

|   | First_Name | Last_Name |
|---|------------|-----------|
| ▶ | Neha       | Sharma    |
|   | Rohan      | AUSTIN    |
|   | Priya      | Kapoor    |
|   | Karan      | Mehta     |

4. Last name = 'AUSTIN'

```
• SELECT *
 FROM EMPLOYEES
 WHERE Last_Name = 'AUSTIN';
```

| Result Grid   Filter Rows:   Edit:   Export/Import:   Wrap Cell Content: |             |            |           |                 |              |            |        |        |                |            |               |
|--------------------------------------------------------------------------|-------------|------------|-----------|-----------------|--------------|------------|--------|--------|----------------|------------|---------------|
|                                                                          | Employee_Id | First_Name | Last_Name | Email           | Phone_Number | Hire_Date  | Job_Id | Salary | Commission_Pct | Manager_Id | Department_Id |
| ▶                                                                        | 102         | Rohan      | AUSTIN    | rohan@gmail.com | 9876543211   | 2022-03-15 | IT     | 7000   | HULL           | 100        | 70            |
| *                                                                        | HULL        | HULL       | HULL      | HULL            | HULL         | HULL       | HULL   | HULL   | HULL           | HULL       | HULL          |

5. Department 60, 70, 80

```
• SELECT First_Name, Last_Name
 FROM EMPLOYEES
 WHERE Department_Id IN (60,70,80);
```

Output:

| Result Grid   Filter Rows: |            |           |
|----------------------------|------------|-----------|
|                            | First_Name | Last_Name |
| ▶                          | Neha       | Sharma    |
|                            | Rohan      | AUSTIN    |
|                            | Aman       | Verma     |
|                            | Priya      | Kapoor    |
|                            | Karan      | Mehta     |

6. Display unique Manager\_Id

```
• SELECT DISTINCT Manager_Id
 FROM EMPLOYEES;
```

Output:

| Result Grid   Filter Rows: |            |
|----------------------------|------------|
|                            | Manager_Id |
| ▶                          | 100        |
|                            | 101        |
|                            | 102        |

## ○ Part C: Updating Records (Client\_master Table)

1. Insert 5 records

- `INSERT INTO Client12 VALUES ('C123','Rohan','Delhi','Delhi',110001,'Delhi',6000);`
- `INSERT INTO Client12 VALUES ('C124','Priya','Mumbai','Mumbai',400001,'Maharashtra',3000);`
- `INSERT INTO Client12 VALUES ('C125','Aman','Jaipur','Jaipur',302001,'Rajasthan',8000);`
- `INSERT INTO Client12 VALUES ('C126','Neha','Lucknow','Lucknow',226001,'UP',2000);`
- `INSERT INTO Client12 VALUES ('C127','Karan','Chandigarh','Chandigarh',160001,'Punjab',9000);`

2. Clients with bal\_due > 5000

- `SELECT Name  
FROM Client12  
WHERE Bal_due > 5000;`

Output:

| Result Grid |       |
|-------------|-------|
|             | Name  |
| ▶           | Rohan |
|             | Aman  |
|             | Karan |

3. Change bal\_due of ClientNO "C123" to 5100

- `UPDATE Client12  
SET Bal_due = 5100  
WHERE ClientNO = 'C123';`

Output:

| Result Grid |          | Filter R |
|-------------|----------|----------|
|             | ClientNO | Bal_due  |
| ▶           | C123     | 5100     |
| *           | NULL     | NULL     |